

ECONOMETRICS I

Problem Set 6: Instrumental variables

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November 23, 2025

1. CRIME AND POLICE: SIMULTANEITY AND IV (SIMULATION)

Consider the following simultaneous equations model that describes the relationship between crime rates and police deployment:

$$\text{Crime} = \alpha_c + \beta_c \text{Police} + \gamma_c \text{Unemp} + \varepsilon_c \quad (\text{eq.C})$$

$$\text{Police} = \alpha_p + \beta_p \text{Crime} + \gamma_p \text{RandBudg} + \varepsilon_p \quad (\text{eq.P})$$

where:

- **Crime** is the crime rate.
- **Police** is police presence.
- **Unemp** is the unemployment rate.
- **RandBudg** is a randomly assigned police budget (think of the government running an experiment).
- ε_c and ε_p are exogenous error terms.

Run the provided R code **crime police** to generate the data and estimate the model. In the code, the true parameter values are: $\beta_c = -2$ (effect of police on crime) and $\beta_p = 1.5$ (effect of crime on police).

- (a) You want to estimate the effect of Police on Crime. Plot Crime vs. Police and explain why ordinary least squares (OLS) regression of Crime on Police would yield biased estimates of β_c .
- (b) Given the equations above, what variable could be used as an instrument to estimate the effect of Police on Crime? Discuss the relevance and exclusion restrictions.
- (c) The manual 2SLS approach involves two stages. Write the equations for both stages and interpret the results from the R output.

- (d) Given the equations above, what variable could be used as an instrument to estimate the effect of Crime on Police? Discuss the relevance and exclusion restrictions.
- (e) Compare the 2SLS estimates with the true parameter values.

2. DISTANCE TO COLLEGE AND RETURN TO EDUCATION (SIMULATION)

Consider the simulation code given in **collegeDistanceSimulation**. It generates the following (artificial) data:

1. Observable variables:

- **educ**: a person's years of education.
- **lwage**: the person's log wage after going to college.
- **urban**: indicates with 1 if the person was raised in an urban center; 0 if raised in a rural place.
- **ldistance**: the log distance to the closest college relative to the living location prior to study.

2. Unobservable processes and variables

- **talent**: represents the talent of a person. It is unobservable but it a variable determining a person's wage.
- **Data generating process (DGP)**:

```
talent <- runif(n_obs,0,10) # Person's talent (unobserved)
urban <- rbinom(n=n_obs, size=1, prob=0.5) # Urban dummy

ldistance <- 1 -urban + rnorm(n_obs) # Log college distance
educ <- 6-gamma*ldistance + talent + rnorm(n_obs) # Years of educ.
educ <- round(educ ,0) # drop decimals

beta_true <- .05 # unknown parameter to be estimated
# log wages:
lwage <- beta_true*educ + 0.5*urban + .02*talent + rnorm(n_obs)
```

Run the code with `n_obs <- 100000` and `gamma <- 1` (seed 1234). You should get the results of table 1.

- (a) Without yet considering the results of the table, why would distance to the nearest college serve in the real world as an instrument for education? Discuss the relevance and exclusion restrictions.
- (b) Now look at the table. You know that the actual coefficient of educ in the DGP is 5%. Why do columns (1) and (2) estimate a much larger value?
- (c) The bias is even larger in (3). Why? Compare to (4)

Table 1: Results with $n = 100,000$ and $\gamma = 1$

	<i>Dependent variable: lwage</i>			
	(1)	(2)	(3)	(4)
	OLS	OLS	IV	IV
educ	0.079*** (0.001)	0.068*** (0.001)	0.151*** (0.003)	0.049*** (0.003)
urban		0.485*** (0.006)		0.503*** (0.007)
Constant	0.039*** (0.011)	-0.080*** (0.011)	-0.713*** (0.032)	0.105*** (0.032)
Observations	100,000	100,000	100,000	100,000
R ²	0.059	0.110	0.011	0.107
Adjusted R ²	0.059	0.110	0.011	0.107
<i>Note:</i>		*p<0.1; **p<0.05; ***p<0.01		

- (d) Rerun the code, now with a smaller values of γ . First try `gamma <- .01` and record your results. Then try `gamma <- .1`. What can you conclude?
- (e) OPTIONAL (because your computer may crash): rerun the whole code keeping the weak instrument (`gamma <- .1`) but with ten million observations (`n_obs <- 10000000`). Now column (4) should again produce an estimate close to 5%.
- (f) Increase the relevance of the instruments by setting $\gamma = 1$, but reduce the number of observations to $n = 1000$. What happened to the estimate in (4)?